

DIGITAL ASSIGNMENT - 3

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1] Given $\Rightarrow T(x, y, z) = (x+2y, x, x-y)$

For $[T]_{\alpha}$, ($\because \alpha = \{c_1, c_2, c_3\}$)

$c_1 = (1, 0, 0)$, $c_2 = (0, 1, 0)$, $c_3 = (0, 0, 1)$

$T(1, 0, 0) = 1(1, 0, 0) + 0(0, 1, 0) + 1(0, 0, 1)$

$T(0, 1, 0) = 2(1, 0, 0) + 0(0, 1, 0) + (-1)(0, 0, 1)$

$T(0, 0, 1) = 0(1, 0, 0) + 1(0, 1, 0) + 0(0, 0, 1)$

$$T_{\alpha} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$$

We know that $[T]_{\beta} = Q^{-1} [T]_{\alpha} Q$

$\beta = \{(1, 1, -1), (1, -1, 1), (-1, 1, 1)\}$

$T(1, 1, -1) = (3, -1, 0)$

$T(1, -1, 1) = (-1, 1, 2)$

$T(-1, 1, 1) = (1, 1, -2)$

$$Q = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 1 & 1 \\ 0 & 2 & -2 \end{bmatrix}$$

$$\begin{aligned} T(1, 0, 0) &= a_1(1, 1, -1) + a_2(1, -1, 1) + a_3(-1, 1, 1) \\ &= a_1 + a_2 - a_3, a_1 - a_2 + a_3, -a_1 + a_2 + a_3 \end{aligned}$$

$\therefore a_1 = 1/2$; $a_2 = 1$; $a_3 = 1/2$

$$T(0,1,0) = (2, 0, -1)$$

$$= (b_1 + b_2 - b_3, b_1 - b_2 + b_3, -b_1 + b_2 + b_3)$$

$$\therefore b_1 = 1 \quad ; \quad b_2 = 1/2 \quad ; \quad b_3 = -1/2$$

$$T(0,0,1) = c_1(1,1,-1) + c_2(1,-1,1) + c_3(-1,1,1)$$

$$\therefore c_1 = 1/2 \quad ; \quad c_2 = 0 \quad ; \quad c_3 = 1/2$$

$$Q^{-1} = \begin{bmatrix} 1/2 & 1 & 1/2 \\ 1 & 1/2 & 0 \\ 1/2 & -1/2 & 1/2 \end{bmatrix}$$

$$[T]_{\beta} = Q^{-1} [T]_{\alpha} Q$$

$$= \begin{bmatrix} 1/2 & 1 & 1/2 \\ 1 & 1/2 & 0 \\ 1/2 & -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 3 & -1 & 1 \\ -1 & 1 & 1 \\ 0 & 2 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 1 & 1/2 \\ 1 & 1/2 & 0 \\ 1/2 & -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 2 & -2 \\ 4 & -2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 5/2 & 3/2 & -1/2 \\ 1 & 2 & 2 \\ 5/2 & -3/2 & 5/2 \end{bmatrix}$$

$$2] \quad (a) \quad \|v_1 - v_2\| = \|(0, 2, -1, 2, 0)\| / \left(\sqrt{0^2 + 2^2 + (-1)^2 + 2^2 + 0^2} \right)$$

$$= \sqrt{9} = 3$$

$$(b) \quad \cos \theta = \frac{\langle v_1, v_2 \rangle}{\|v_1\| \cdot \|v_2\|} = \frac{1 - 1 + 0 - 1 + 1}{\sqrt{4} \times \sqrt{5}} = 0$$

$$\cos \theta = 0 \Rightarrow \theta = \cos^{-1} 0 = \pi/2$$

$$(c) \text{proj}_{v_1} = \frac{\langle v_1, v_2 \rangle}{\langle v_2, v_2 \rangle} \cdot v_2 = 0, 0, 0$$

$$\text{proj}_{v_2} = \frac{\langle v_1, v_2 \rangle}{\langle v_1, v_1 \rangle} \cdot v_1 = 0, 0, 0$$

$$3] \quad A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & -1 \\ -1 & 2 & 1 \end{bmatrix}$$

$$n_1 = 1, 1, 1, -1$$

$$n_2 = 2, 1, 0, 2$$

$$n_3 = -1, 1, -1, 1$$

$$v_1 = n_1 = 1, 1, 1, -1$$

$$v_2 = n_2 - \frac{\langle n_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 = (2, 1, 0, 2) - \frac{1}{4} (1, 1, 1, -1)$$

$$v_3 = n_3 - \frac{\langle n_3, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 - \frac{\langle n_3, v_2 \rangle}{\langle v_2, v_2 \rangle} v_2$$

$$= (-1, 1, -1, 1) + \frac{1}{2} (1, 1, 1, -1) - \frac{6}{35} \left(\frac{7}{4}, \frac{3}{4}, -\frac{1}{4}, \frac{9}{4} \right)$$

$$= \left(-\frac{4}{5}, \frac{4}{5}, -\frac{11}{35}, \frac{4}{35} \right)$$

$$c_1 = \frac{v_1}{\|v_1\|} = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2} \right)$$

$$c_2 = \left(\frac{7}{2\sqrt{35}}, \frac{3}{2\sqrt{35}}, \frac{-1}{2\sqrt{35}}, \frac{9}{2\sqrt{35}} \right)$$

$$c_3 = \left(\frac{-7}{\sqrt{210}}, \frac{12}{\sqrt{210}}, \frac{-4}{\sqrt{210}}, \frac{1}{\sqrt{210}} \right)$$

$$Q = \begin{bmatrix} 1/2 & 7/2\sqrt{35} & -7/\sqrt{210} \\ 1/2 & 3/2\sqrt{35} & 12/\sqrt{210} \\ 1/2 & -1/2\sqrt{35} & -4/\sqrt{210} \\ 1/2 & 9/2\sqrt{35} & 1/\sqrt{210} \end{bmatrix}$$

$$\pi_{11} = \langle n_1, c_1 \rangle$$

$$= 1/2 + 1/2 + 1/2 + 1/2 = \underline{\underline{2}}$$

$$\pi_{12} = \langle n_2, c_1 \rangle$$

$$= 1 + 1/2 + 0 - 1 = \underline{\underline{1/2}}$$

$$\pi_{13} = \langle n_3, c_1 \rangle$$

$$= -1/2 + 1/2 - 1/2 - 1/2 = \underline{\underline{-1}}$$

$$\pi_{21} = \underline{\underline{0}}$$

$$\pi_{22} = \langle n_2, c_2 \rangle = \frac{\sqrt{35}}{2} \underline{\underline{}}$$

$$\pi_{23} = \langle n_3, c_2 \rangle = \underline{\underline{3/\sqrt{35}}}$$

$$\pi_{31} = 0$$

$$\pi_{32} = 0$$

$$\pi_{33} = \langle n_3, c_3 \rangle = \underline{\underline{24/\sqrt{210}}}$$

$$\Rightarrow R = \begin{bmatrix} 2 & 1/2 & -1 \\ 0 & \sqrt{35}/2 & 3/\sqrt{35} \\ 0 & 0 & 24/\sqrt{210} \end{bmatrix}$$

$$\Rightarrow Q = \begin{bmatrix} 1/2 & 7/2\sqrt{35} & 57/\sqrt{210} \\ 1/2 & 3/2\sqrt{35} & 12/\sqrt{210} \\ 1/2 & -2/2\sqrt{35} & -4/\sqrt{210} \\ 1/2 & 9/2\sqrt{35} & 1/\sqrt{210} \end{bmatrix}$$

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